

4(a)	any five from: 1 (named) <u>restriction</u>, enzyme / endonuclease ; 2 cut / fragment, DNA ; 3 produced by / present in, bacteria / archaea / prokaryotes or - to defend against, viruses / (bacterio)phages ; 4 (DNA) ligase ; 5 joins DNA / forms phosphodiester bonds / seals sugar-phosphate backbone ; 6 during DNA, repair / replication or - joins Okazaki fragments ; 7 reverse transcriptase ; 8 forms / makes, (c)DNA from mRNA (template) ; 9 produced by / present in, retroviruses ; 10 DNA / <i>Taq</i>, polymerase ; 11 replicates / copies, DNA (at high temperatures) ; 12 produced by / present in, <i>Thermus aquaticus</i> ;	5
4(b)(i)	any three from: 1 increased / higher / more, yield / production / harvest / food ; 2 increased / higher / better, quality of, produce / crops / food ; 3 decreased / lower / less, insecticide / pesticide, needed / used ; 4 decreased / lower / less, fuel use / CO₂ emissions ; 5 potential for, set-aside / conservation areas (as less land needed for crops) ; 6 (described) comparative economic benefit ;	3
4(b)(ii)	any two from: 1 people in other (named) continents eat GM, food / crops ; 2 GM, food / crops, already in (European) food chain ; 3 <i>ref. to</i> economic benefit of reducing GM (animal food) imports ;	2
4(b)(iii)	1 do not know / need to know, total crop area / non-GM crop area ; 2 so can work out, percentage / fraction, of, GM / non-GM ;	2